Multiphysics 3-D optimisation of electric motors on pretrained surrogates

A large physics model — built once, shipped with its error bars — that replaces the FEA and the thermal solve inside the design loop, so a motor can be optimised over thousands of designs without a single finite-element call.

PINNeAPPle · `pinneapple_simulation.electromagnetics.machines.motor3d` · every number in this document was produced by the code it describes, on a laptop CPU, with the artefacts kept next to it.

1. The question this document answers

An electric motor is the canonical multiphysics object. The stator current produces torque *and* heat; the heat raises the winding and magnet temperatures; the magnet temperature lowers the remanence, which lowers the torque and eats the demagnetisation margin; the coolant flow sets how much heat leaves; the stack length and the skew change all of the above at once. Every one of those couplings is real, and every one of them is expensive to evaluate, because the electromagnetic side is a nonlinear magnetostatic finite-element problem that must be solved at several rotor positions per operating point.

The engineering consequence is that motor design still lives with a small number of FEA runs and a large amount of experience. The alternative explored here is to spend the FEA once — on a designed set of cases — and learn the *whole chain* from geometry and operating point to performance, loss, temperature and demagnetisation margin. The learned model is not a curve fit per quantity: it is one map from the same eleven design variables to the fourteen physics outputs plus three axial profiles, shipped as a pretrained artefact with a model card that states its held-out error on every one of them.

> **What "without FEA or CFD" means here, precisely.** During the search, every candidate design is answered by a forward pass of an ensemble of small networks — about 100 microseconds per design, and no solver involves. The FEA is spent twice: once, offline, to build the training set, and once, at the end, on a handful of designs of the Pareto front, to check that the front is real. Both costs are reported in every result. Nothing in this document claims a neural network knows physics it was not shown.

2. What the repository already contained

Before this work the library had most of the physics and all of the optimisation scaffolding, but not the loop that connects them. The inventory matters, because the model added here is built out of these pieces rather than next to them.

Capability	Where	What it does
2-D nonlinear magnetostatic FEA of a machine cross-section	`machines.IPMMachine`, `machines.prius_2004`	Torque by Maxwell stress and by co-energy, flux linkage, back-EMF, `L_d` / `L_q`, MTPA, iron and magnet loss — the 2004 Toyota Prius geometry, verified against published FEA.
Conforming air-gap mesher with a moving band	`machines.mesher`	One pole and one slot pitch meshed with every polygon edge as a mesh edge, then copied around the circle.
Catalogue steel and magnet grades	`machines.steel`	B–H curve and loss table of M400-50A, NdFeB with its reversible `Br(T)` coefficient.
End windings outside the cross-section	`machines.endwinding`, `machines.drive`	End-turn length, resistance and leakage of the coil ends; flux maps and drive-cycle energy.
Lumped thermal network, one node per part	`machines.thermal.machine_network`	Seven nodes (yoke, teeth, winding, end windings, rotor, magnets, shaft) and a demagnetisation margin against `Hcj(T)`.
Design-space optimisation with a surrogate	`pinneapple_design.design_optimizer`	NSGA-II, gradient methods, a Gaussian-process surrogate with active learning, hypervolume, and re-verification of the front by the reference solver.
Magnetic topology optimisation	`pinneapple_design.topology.magnetic`	Rotor material distribution against mean torque.
An industrial preset for the Prius motor	`pp.presets.load("ipm_traction_motor")`	Torque, ripple, MTPA and voltage with a data sheet, a surrogate built on demand, an inversion and a validation against Yang et al. (2013).
The 3-D surrogate pattern, proved on another domain	`pinneapple_design.aero` (aircraft)	MLP coefficients over airfoil + planform, a whole-aircraft evaluator on top, NSGA-II, trust from the ensemble spread, ONNX export for geometry.

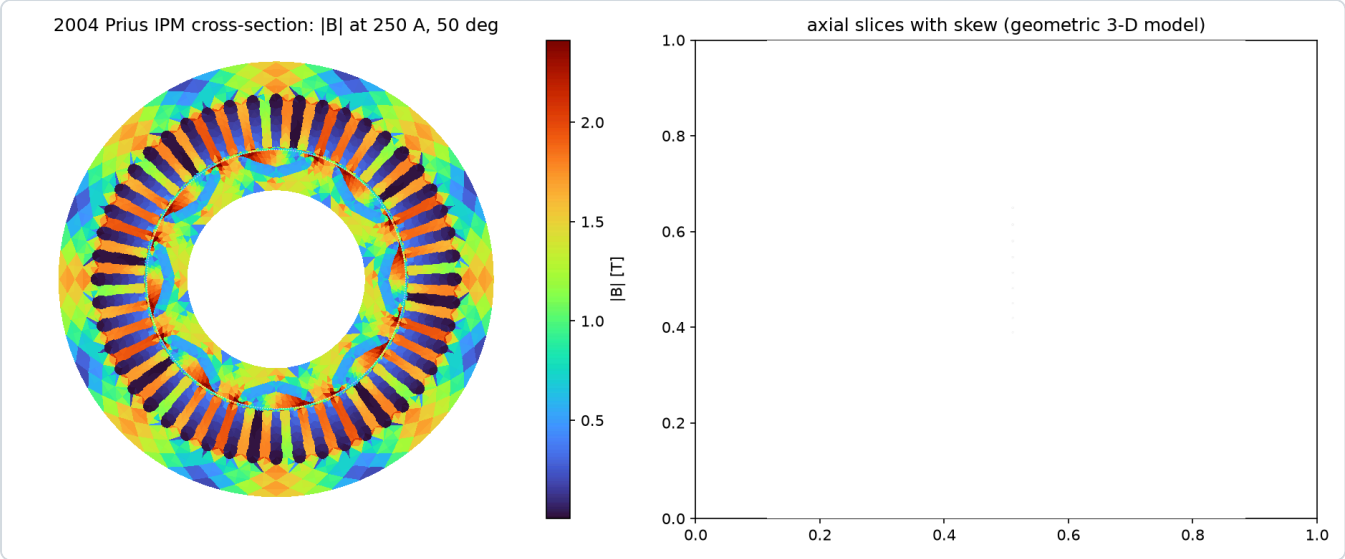
Model zoo, trust layer, benchmarks	`pp.zoo`, `pp.validate`, `pinneapple check`	Pretrained models with SHA-256-checked weights and cards; physics audit of any prediction; equal-budget benchmarks.

Two gaps followed from that inventory. First, the machine model was strictly two-dimensional per cross-section and the thermal network had one node per part: enough to know a hot-spot *level*, not its *place* along a stack with two cooled ends. Second, nothing assembled the chain — FEA, skew, loss, cooling, the magnet temperature back on torque and demagnetisation — into one call, and nothing in the motor domain had a pretrained surrogate that could be evaluated a hundred thousand times.

3. What was added

One package, `motor3d`, with six modules, a pretrained artefact, tests, an example and a guide:

Module	Contribution
`thermal3d.py`	The radial thermal network cut into axial segments and linked by `k A / dz`, with the end windings as one node outside the stack and the water jacket spread over every yoke segment. With one segment it reproduces `machines.thermal.machine_network` to round-off, which is the test that says it is the same physics and not a second opinion.
`model.py`	The coupled chain in one call: FEA at the operating point, skew by axial slices, iron/magnet/copper loss, cooling at a coolant flow, the magnet temperature corrected back onto torque, iron loss and the demagnetisation margin, and the axial profiles. Designs past the reversible range of the magnet grade are reported as not assessable instead of scored.
`design.py`	Eleven design variables declared once — rotor cross-section, stack, skew, operating point, coolant flow — with units and bounds. The design space, the dataset columns and the network inputs are all derived from that single declaration.
`dataset.py`	A design of experiments over that box with the geometry test as a feasibility filter, evaluated in parallel processes and stored with its provenance (method, seed, model configuration, cost, rejection rate).
`surrogate.py`	The ensemble: five to eight networks, log-space training for strictly positive outputs, a held-out error per output, a validity box, an extrapolation measure, constant-output detection, and SHA-256-checked weights with a model card.
`optimize.py`	The design problem and the NSGA-II search that runs entirely on the surrogate, the Pareto front, and the re-check of the front with the reference model. Constraints are posed on the *conservative* value (mean + 1σ of the ensemble), so a design that only satisfies a limit inside the model does not enter the front.
`workflow.py`	Data → train → optimise → verify → figures → `report.md`, as one command with a subcommand per step.



Left: the reference field solve — |B| in the 2004 Prius cross-section at 250 A peak and 50° current angle, from the FEA the surrogate learns from. **Right:** the axial model: the stack cut into slices, each twisted by the design's skew, which is how a skewed machine is represented without a 3-D field solve. Both figures are produced by `Motor3DModel.solve(..., keep_fields=True)`.

4. The physics chain

4.1 Electromagnetic, per axial slice

One call to `IPMMachine.operating_point` solves the nonlinear magnetostatic problem $-\mathrm{div}(\nu\,\nabla A_z) = J + \mathrm{curl}(\nu\,B_r)$ over a set of rotor angles with sinusoidal currents fixed in the dq frame, and returns torque per angle, ripple, ψ_d, ψ_q and the fields. The magnet remanence is evaluated at the magnet temperature the design currently has, not at 20 °C.

4.2 Skew as slice averaging

The stack is cut into `n_segments` slices, each at its own electrical angle $\delta_k = (k/(K-1) - \tfrac{1}{2})\cdot\mathrm{skew}\cdot p$. Each slice contributes the torque of the cross-section at that angle, so slot harmonics cancel exactly as they do in a skewed machine, and the ripple of the skewed stack is computed from the concatenated slice waveforms. The fundamental is reduced by the classic skew factor $\sin(x)/x$ with $x = p\cdot\mathrm{skew}/2$. That the mean torque of a 2-D slice model is otherwise invariant to the twist is a property of the model, not an oversight: the axial flux paths that skew also perturbs enter here through the end-winding resistance and the thermal network, and nowhere else.

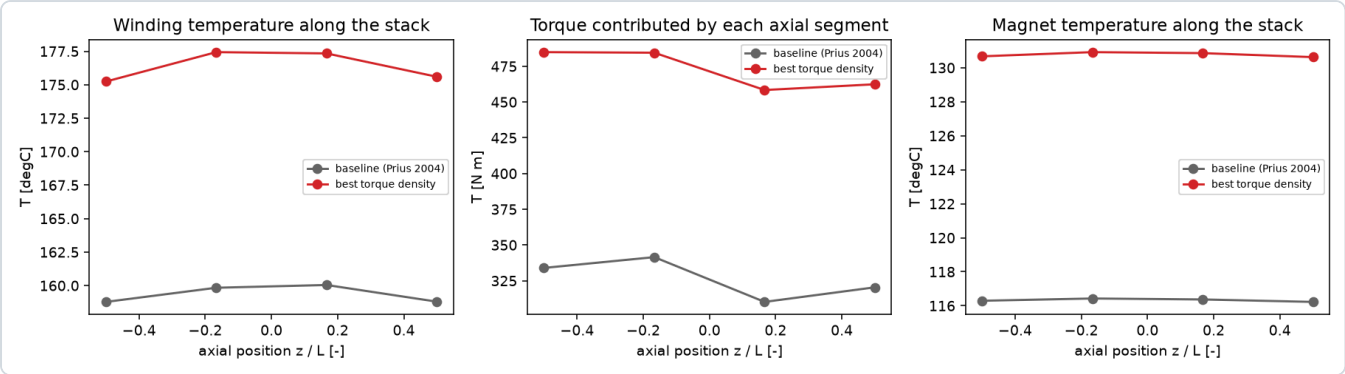
4.3 Losses, with the winding's axial geometry

Iron loss comes from the flux density over an electrical period, per element, in the element's own frame; magnet loss from the resistance-limited eddy model; copper loss from the phase resistance, which depends on the stack length and on the end-turn length computed from the coil pitch at the mid-slot radius — a genuinely 3-D quantity. The copper loss is split between the slot and the end windings in proportion to their lengths, because the two sit in different nodes of the thermal network.

4.4 Cooling and the axial thermal network

The jacket conductance follows the coolant flow as $\mathrm{flow}^{0.8}$ — turbulent convection, stated as an assumption rather than fitted. The network solves $(L + G_b)T = P + G_b T_\infty$ on 5·K + 1 nodes. Its value is visible in the figure

below: the winding is hottest in the middle of the stack and cooler at the ends that face the end windings, which is a statement a one-node-per-part model cannot make.



Axial resolution, baseline Prius design against the best torque-density design of the front, as predicted by the pretrained surrogate. Left: winding temperature along the stack. Middle: the torque contributed by each axial segment at a fixed rotor angle — the skew redistributes it. Right: magnet temperature. These three curves are outputs of the surrogate, not post-processing.

4.5 The coupling back to the magnets

The magnet temperature changes the remanence, and the remanence changes torque, iron loss (through B^2) and the demagnetisation margin. The reference model solves the field once at a first-guess magnet temperature, solves the network, corrects those three quantities analytically and re-evaluates the copper loss at the resulting winding temperature until the temperatures stop moving. That is a one-step fixed point around one FEA solve per design; setting `fea_iterations=2` re-solves the field at the converged temperature, and the difference between the two is measured rather than assumed. The demagnetisation check compares the worst reverse field inside every magnet element against the intrinsic coercivity of the grade at that temperature; designs whose magnet would be pushed past the reversible range of the grade are reported as not assessable, because extrapolating $H_{cj}(T)$ through zero would produce a number with no meaning.

5. Building the large model

5.1 The design vector

```
from pinneapple_simulation.electromagnetics.machines.motor3d import Motor3DDesign, VARIABLES

for v in VARIABLES:
    print(f"{v.name:12s} {v.unit:8s} [{v.low}, {v.high}]  {v.description}")

d = Motor3DDesign.baseline()          # the 2004 Prius at its published operating point
d = Motor3DDesign(L=0.10, skew_deg=6.0, rpm=3000.0, flow_L_min=8.0)
d.sector_errors()                     # [] when the cross-section can be meshed
```

The eleven variables are `W4`, `H3`, `H0`, `W0`, `H1` (rotor cross-section, the same parametrisation the 2-D studies use), `L` and `skew_deg` (axial), `I_peak`, `gamma_deg`, `rpm` (operating point) and `flow_L_min` (cooling). Every one of them changes more than one output, which is the point.

5.2 The reference model in one call

```
from pinneapple_simulation.electromagnetics.machines.motor3d import Motor3DModel, PRIUS

model = Motor3DModel()                # 192-node gap mesh, 12 rotor angles, 4 axial segments
r = model.solve(PRIUS)                # 250 A, 50 deg, 1500 rpm, 6 L/min coolant
print(r["torque_Nm"], r["T_winding_C"], r["demag_ratio"])
print(r["T_winding_profile_C"])   # the winding along the stack, end to end
print(r["extra"]["v_phase_V"], r["extra"]["v_limit_V"])
```

A design whose cross-section the mesher refuses, or whose magnet would demagnetise, returns `valid=False` with the reason and no solve: the reference model is allowed to say "this cannot be assessed", and the dataset keeps that answer.

5.3 The training set

```
from pinneapple_simulation.electromagnetics.machines.motor3d import build_dataset, Motor3DDesign

ds = build_dataset(1200, seed=20260101, workers=8,
                   model=Motor3DModel(n_gap=192, n_positions=12), extra=
[Motor3DDesign.baseline()])
ds.save("motor3d_data")
print(ds.meta["rejection_rate"], ds.meta["seconds_per_design"])
```

Designs drawn (Latin hypercube, geometry filter applied)	1201
Buildable and solved	967 (rejection rate 19.5 %)
Reference cost on 8 cores	384.5 s wall, 0.32 s per design, 981 FEA operating points
FEA configuration	192 gap nodes × 12 rotor angles, 4 axial segments, one field iteration

The rejection rate is itself a result: one design in five drawn uniformly from the declared box cannot be built — polygons that cross, a magnet that leaves its pole. An optimiser that does not know this spends a fifth of its budget on geometry the mesher refuses, which is why the exact geometry test stays inside the search loop.

5.4 The surrogate, and what its card says

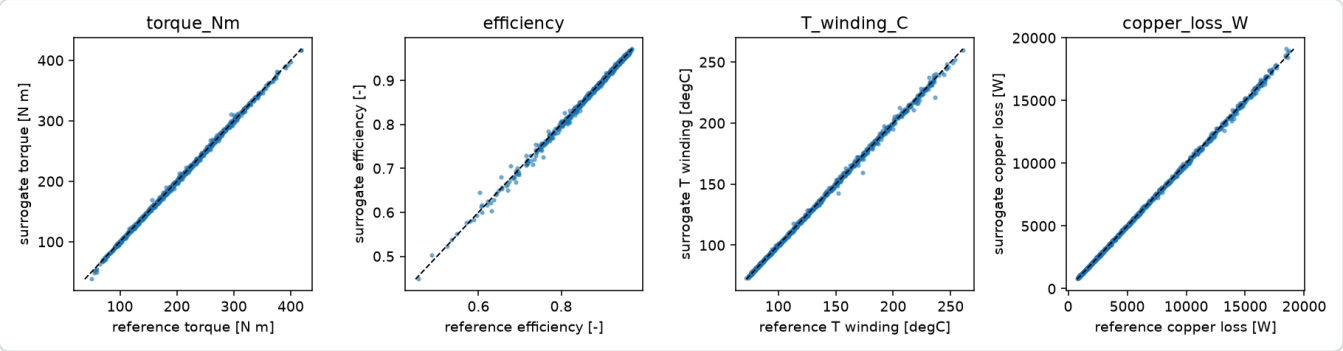
```
from pinneapple_simulation.electromagnetics.machines.motor3d import MotorSurrogate3D,
SurrogateConfig

surrogate = MotorSurrogate3D(SurrogateConfig(ensemble=8, epochs=3000, seed=7)).fit(ds)
card = surrogate.card()
print(card["heldout"]["torque_Nm"])          # mae, r2, mean_rel, p95_rel
print(card["mean_r2"], card["worst_output"])
surrogate.save("motor3d-v1", version="motor3d-v1")
```

Each network maps the eleven normalised design variables to 25 targets: the fourteen scalars (torque, ripple, torque density, efficiency, three loss channels, three temperatures, demagnetisation ratio, two masses) and three axial profiles of four segments. Outputs that are strictly positive are trained in log space, because what matters about a 300 W loss and a 30 N·m torque is their relative error, not their absolute one. An output that does not move in the training set is detected and predicted as its constant instead of being learned — that is how the magnet eddy loss, which is zero at these speeds with solid magnets, is reported rather than modelled as noise.

Held-out output (193 designs never trained on)	MAE	R^2	mean rel.	95th rel.
Torque [N·m]	2.07	0.998	1.15 %	2.88 %
Torque density [N·m/kg]	0.054	0.998	1.08 %	2.65 %
Efficiency [–]	0.0037	0.992	0.48 %	1.88 %
Copper loss [W]	82	0.999	1.51 %	4.21 %
Iron loss [W]	35	0.995	4.04 %	12.99 %
Winding temperature [°C]	1.39	0.997	0.92 %	2.53 %
Magnet temperature [°C]	1.05	0.996	0.87 %	2.44 %
Demagnetisation ratio [–]	0.020	0.989	3.10 %	8.20 %
Torque ripple [–]	0.014	0.788	8.86 %	20.50 %
mean R^2 over the outputs that vary		0.983		

Ripple is the honest weak spot: `R² = 0.79`, with a 95th-percentile relative error of 20 %. It is the output that depends most on the fine structure of the slot geometry, and the one the constraints are most likely to press against. That is the reason the constraint is not posed on the network's mean — see the next section.



Surrogate against the reference model on 967 buildable designs (colour = density). Torque, efficiency, winding temperature and copper loss are reproduced to about one per cent; the deviations grow where the machine saturates, which is where the reference model's own mesh matters.

6. The optimisation, with no solver in the loop

```
from pinneapple_simulation.electromagnetics.machines.motor3d import load_pretrained,
design_problem, optimize, verify_front

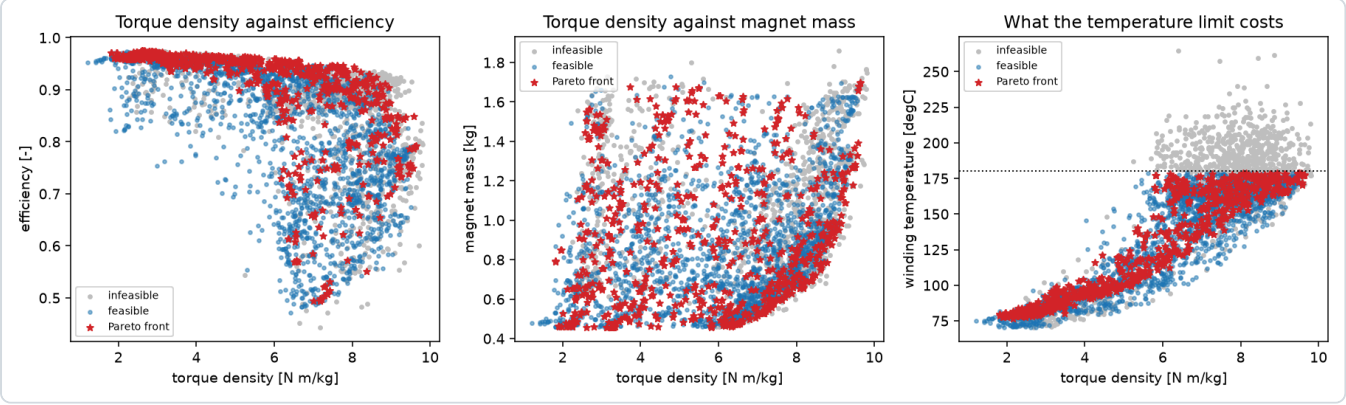
surrogate = load_pretrained()                       # weights checked against the SHA-256 of the
card
out = optimize(design_problem(surrogate=surrogate), budget=4000, seed=0)
print(out["n_evaluations"], out["designs_per_second"], out["n_reference_solves"])
ver = verify_front(out, n=8)                        # the only place an FEA call is spent
print(ver["still_feasible"], ver["rank_agreement"], ver["errors"]["torque_Nm"])
```

The problem is the one a motor designer asks: maximise torque density and efficiency, minimise magnet mass, subject to the winding temperature (180 °C, class H), the magnet temperature (140 °C, where the coercivity of a sintered NdFeB starts to fall away), the demagnetisation ratio (below the knee, i.e. 1) and the torque ripple

(20 % peak to peak, which a traction inverter and gearbox tolerate). The search is NSGA-II from `pinneapple_design.design_optimizer`, and every candidate it evaluates is a network forward pass.

> **Two details that make the difference between a front and a plausible-looking scatter.**
> 1. *Constraints on the conservative value.* The temperature, demagnetisation and ripple limits are imposed on `mean + 1σ` of the ensemble, not on its mean. A design at the limit is exactly where a one-sigma optimistic prediction is worthless, and the first version of this study — constraints on the mean — produced a front in which five of eight re-checked designs failed the same limits. With the conservative form, eight of eight pass.
> 2. *The geometry test stays exact.* The surrogate maps numbers to numbers; it does not know whether a cross-section can be built. `sector_errors()` costs 2.9 ms, so it runs inside the evaluator and marks unbuildable designs invalid instead of being learned.

Run of the study (as shipped in `report.md`)	Value
Designs evaluated by the surrogate	4000
Wall time of the search	28.7 s
Throughput, geometry test included	139 designs/s
FEA calls made by the search	0
Designs on the Pareto front	680
Best torque density	9.64 N·m/kg (442 N·m, magnet 1.70 kg, winding 177 °C, ripple 0.168)
Best efficiency	0.976 (123 N·m, winding 84 °C)
Reference model, per design	1.34 s
Surrogate, per design	99 µs
Ratio	13 500×



The front over four thousand designs, with the two limits that bind: the winding temperature and the demagnetisation margin. The rightmost panel is the most informative — the temperature limit is what stops torque density at about 10 N·m/kg, and that is a statement about the cooling, not about the electromagnetics.

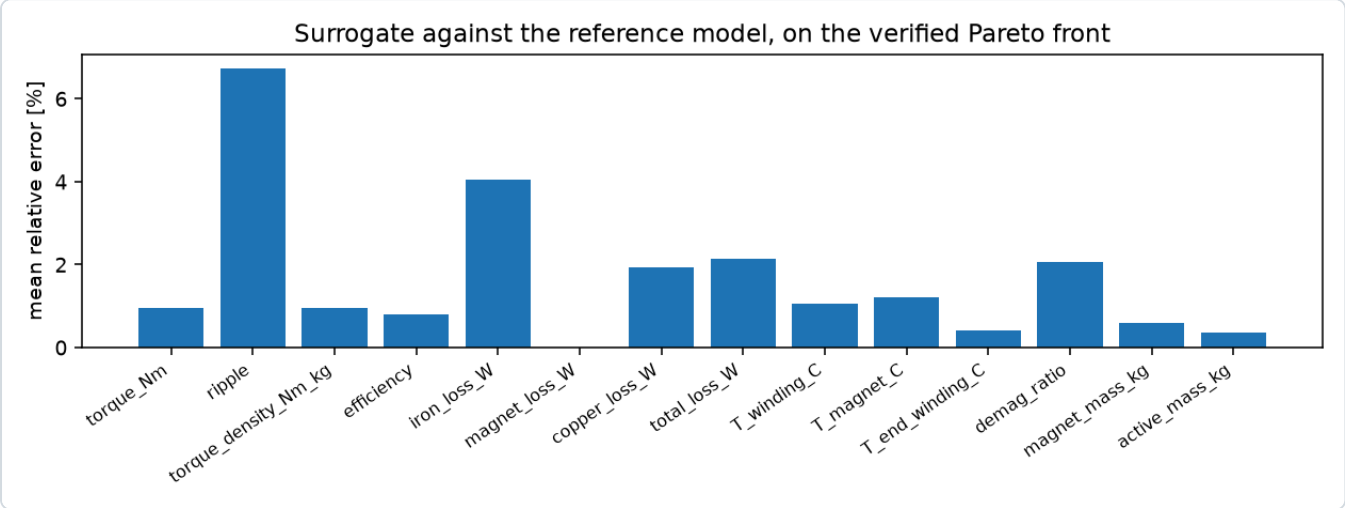
7. Checking the front against the physics

```
import json
from pathlib import Path
from pinneapple_simulation.electromagnetics.machines.motor3d from ... # see workflow._jsonable
ver = verify_front(out, reference=Motor3DModel(), n=8)
```

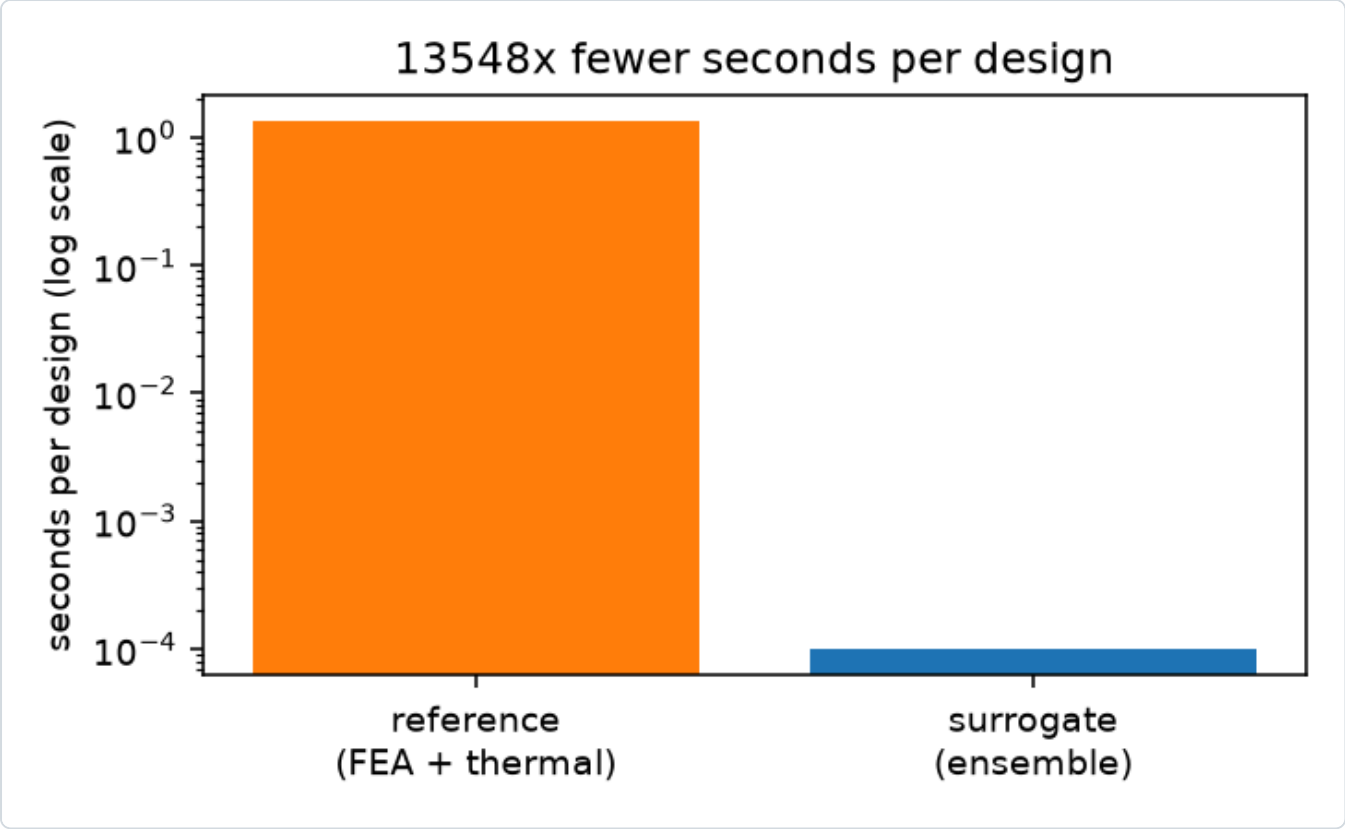
Eight designs spread across the front by farthest-point sampling are re-evaluated by the reference model: 16.3 s of FEA, against the 28.7 s the surrogate search itself took. The result is a statement about the front, which is the region the optimiser actually chose — not about a random split.

Output	mean rel. error on the front	max rel. error
Torque	0.96 %	2.54 %
Torque density	0.95 %	1.99 %
Winding temperature	1.06 %	1.83 %
Demagnetisation ratio	2.05 %	5.39 %
Efficiency	0.81 %	5.10 %
Torque ripple	6.73 %	20.64 %

8 of 8 designs of the front remain feasible when the reference model evaluates them under the same limits, and the Spearman rank agreement on torque density between surrogate and reference is **1.000** — the front is ordered identically. The residual spread in ripple is the same number the model card reports, and it is the number to keep in mind when the ripple limit is the binding constraint of a study.



Surrogate against the reference model, on the verified front. The ripple bar is the model's known weakness; everything the front was actually optimised for sits below a few per cent.



One reference design and one surrogate design on the same machine in the same process: 1.34 s against 99 µs. The break-even point is around 300 designs — a training set of a thousand designs pays for itself in the first optimiser run, and every later run on that design space is free.

8. What the study does not model

Read this before quoting a number from the previous pages.

- **The field is 2-D per axial slice.** Axial flux paths — end leakage, axial saturation — enter through the end-winding resistance and the thermal network, not through a 3-D field solve. A full 3-D FE motor model is not part of this pipeline, and nothing here is a substitute for one when the question is genuinely axial.
- **The coolant is a boundary conductance.** The heat-transfer coefficient follows the flow as $\texttt{flow}^{0.8}$; it is an input, not the output of a conjugate heat transfer solution. A validated cooling model needs coefficients from measurements or from a CFD study of the actual jacket.
- **The structural side is absent.** Bridge stress at overspeed, magnet retention, rotor dynamics: none of it is evaluated. A design that survives this study has passed a thermal and electromagnetic test only.
- **The surrogate is valid inside its box.** The ensemble spread is a proxy for uncertainty, not a calibrated interval; `out_of_box` and `extrapolation` say when a design has left the data. Beyond the box, the card's errors no longer apply.
- **Magnet eddy loss is zero at the speeds studied** with solid magnets; it is detected as a constant output rather than modelled. Segmenting the magnets gives a non-zero, resistance-limited value.
- **The reference itself is a model.** Its torque agrees with the published FEA of Yang et al. (2013) to a few per cent through `machines.verification`, and the axial network reduces to the library's seven-node network exactly — but those are consistency checks against models, not against a bench test.

9. Reproducing it

```
# the whole study: dataset, surrogate, search, verification, figures, report.md
python -m pinneapple_simulation.electromagnetics.machines.motor3d.workflow all --out motor3d_run

# or step by step
python -m ...workflow build-dataset --n 1200 --workers 8 --out motor3d_run/dataset
python -m ...workflow train        --data motor3d_run/dataset --out motor3d_run/model
python -m ...workflow optimize     --budget 4000 --out motor3d_run/optimization.json
python -m ...workflow verify       --optimization motor3d_run/optimization.json --n 8
python -m ...workflow check-coupling --n 4     # cost of the one-step Br(T) correction

# the worked example, with the numbers printed at every step
python examples/motor_magnetics/03_multiphysics3d_surrogate.py
```

The test suite covers the physics reduction (one segment = the library's seven-node network, to round-off), the direction of every coupling (more skew → less torque and less ripple; more coolant → colder magnets → more torque), the deliverability of the training set, the model card, the SHA-256 check that refuses tampered weights, the search with its zero-FEA claim, the verification path, and the pretrained artefact itself.

Appendix A — files

Path	Content
`pinneapple_simulation/electromagnetics/machines/motor3d/design.py`	Design vector, machine geometry, skew factor
`.../motor3d/thermal3d.py`	Axial-resolved lumped thermal network
`.../motor3d/model.py`	The coupled multiphysics reference model
`.../motor3d/dataset.py`	DOE, parallel evaluation, stored provenance
`.../motor3d/surrogate.py`	Ensemble, model card, SHA-256 loading
`.../motor3d/optimize.py`	Design problem, NSGA-II on the surrogate, verification
`.../motor3d/workflow.py`	CLI, figures, `report.md`
`.../motor3d/data/motor3d-v1/`	Pretrained weights (1.2 MB) and model card
`.../motor3d/data/train_v1/`	The training set with its metadata
`tests/test_motor3d.py`	16 tests
`examples/motor_magnetics/03_multiphysics3d_surrogate.py`	Worked example
`docs/guides/motor3d_surrogate.md`	The guide

Appendix B — the public API

```
Motor3DDesign, VARIABLES, NAMES, BOUNDS       # the design vector, declared once
Motor3DModel(n_gap, n_positions, n_segments)  # .solve(design) -> every quantity
                                              # .evaluate(X)   -> arrays over designs
AxialThermalModel(geometry, data, n_segments) # .steady(P), .powers(...)
build_dataset(n, seed, workers, model)        # -> MotorDataset (.save / .load)
MotorSurrogate3D(config).fit(dataset)         # .predict, .card, .save, .load
load_pretrained()                             # the shipped model, hash-checked
design_problem(surrogate=...)                 # objectives + limits + design space
optimize(problem, budget, seed)               # -> front, best designs, cost
verify_front(result, reference, n)            # the FEA re-check of the front
benchmark(surrogate, reference, n)            # seconds per design, both models
```

PINNeAPPle — physics AI that tells you when to trust it. This document and the artefacts it describes are part of the repository at `pinneapple_simulation/electromagnetics/machines/motor3d`.